21. Find the Union of Two Arrays

Description

Given two arrays, find all distinct elements present in either array.

Input Format

First line: N

Second line: N integers

Third line: M

Fourth line: M integers

Sample Input

5

1 2 3 4 5

4

1 2 6 7

Sample Output

1 2 3 4 5 6 7

Explanation

Union contains all unique elements from both arrays.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n,m;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
```

```
    }
    scanf("%d",&m);
    int arr2[m];
    for(int i=0;i<m;i++){
        scanf("%d",&arr2[i]);
    }
    for(int i=0;i<n;i++){
        printf("%d ",arr[i]);
    }
    int k=0;
    for(int i=0;i<m;i++){
        for(int j=0;j<n;j++){
          if(arr2[i]==arr[j]){
            k++;
          }
        }if(k==0){
            printf("%d ",arr2[i]);
        }
        k=0;
    }


    return 0;
}
```

22. Find the Intersection of Two Arrays

Description

Find common elements present in both arrays.

Sample Input

5

1 2 3 4 5

5

3 4 5 6 7

Sample Output

3 4 5

Explanation

3, 4, and 5 are present in both arrays.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n,m;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    scanf("%d",&m);
    int arr2[m];
    for(int i=0;i<m;i++){
        scanf("%d",&arr2[i]);
    }
    int k=0;
    for(int i=0;i<m;i++){
        for(int j=0;j<n;j++){
          if(arr2[i]==arr[j]){
            k++;
```

```
        }
      }if(k>0){
         printf("%d ",arr2[i]);
      }
      k=0;
   }



   return 0;
}
```

23. Find the First Repeating Element

Description

Find the first element that appears more than once.

Sample Input

7

10 5 3 4 3 5 6

Sample Output

5

Explanation

5 is the first element whose repetition occurs earliest.

PROGRAM:

```
#include <stdio.h>

int main()
{
   int n;
   scanf("%d",&n);
```

```
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }

    int i=0,j=1;
    while(i<j){
        if(j<n && arr[i]==arr[j]){
            printf("First Repeating Element %d",arr[i]);
            return 0;
        }j++;
        if(j==n-1){
            i++;
            j=i+1;
        }
    }

    return 0;
}
```

24. Find the First Non-Repeating Element

Description

Find the first element that occurs exactly once.

Sample Input

6

9 4 9 6 7 4

Sample Output

6

Explanation

6 is the first element with frequency 1.

PROGRAM:

```
#include <stdio.h>

int main()
{
   int n;
   scanf("%d",&n);
   int arr[n];
   for(int i=0;i<n;i++){
      scanf("%d",&arr[i]);
   }
   int copy[10000];
   for(int i=0;i<n;i++){
      copy[arr[i]]=0;
   }
   for(int i=0;i<n;i++){
      copy[arr[i]]++;
   }
   for(int i=0;i<n;i++){
      if(copy[arr[i]]==1){
         printf("%d ",arr[i]);
         break;
      }
   }



   return 0;
}
```

25. Maximum Consecutive Ones

Description

Find the maximum number of consecutive 1s in a binary array.

Sample Input

8

1 1 0 1 1 1 0 1

Sample Output

3

Explanation

The longest sequence of consecutive 1s has length 3.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int s=0;s<n;s++){
        scanf("%d",&arr[s]);
    }
    int max=0,c=0;
    for(int i=0;i<n;i++){
        if(arr[i]==1){
            c++;
        }
        if(c>max){
```

```
            max=c;
        }
        if(arr[i]==0){
            c=0;
        }
    }printf("number of consecutive 1s in a binary array %d",max);

    return 0;
}
```

26. Find All Duplicate Elements

Description

Print all elements that appear more than once.

Sample Input

8

1 2 3 1 2 4 5 2

Sample Output

1 2

Explanation

Elements 1 and 2 occur multiple times.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n;
```

```
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int k=0;
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                k++;
            }
        }if(k>0){
            printf("%d ",arr[i]);
        }k=0;
    }


    return 0;
}
```

27. Largest Sum Contiguous Subarray of Size K

Description

Find the maximum sum among all subarrays of size K.

Sample Input

6

2 1 5 1 3 2

3

Sample Output

9

Explanation

Subarray [5,1,3] has sum 9.

PROGRAM:

```
#include <stdio.h>

int main()
{
   int n,k,sum=0,maxsum=0;
   scanf("%d",&n);
   int arr[n];
   for(int s=0;s<n;s++){
      scanf("%d",&arr[s]);
   }
   scanf("%d",&k);
   for(int i=0;i<k;i++){
      sum+=arr[i];
   }
   maxsum=sum;
   int y=k;
   for(int i=k;i<n;i++){
      int x=(maxsum-arr[i-k])+arr[y];
      y++;
      maxsum=x;
      if(x>sum){
         sum=x;
      }
   }
   printf("largest sum contigious subarray of k is %d",sum);

   return 0;
```

```
}
```

28. Find the Smallest Positive Missing Number

Description

Find the smallest positive integer missing from the array.

Sample Input

5

3 4 -1 1 2

Sample Output

5

Explanation

Numbers 1,2,3,4 exist, so 5 is missing.

PROGRAM:

```
#include <stdio.h>

#include<stdlib.h>

int comp(const void *a,const void *b){

   return(*(int*)a - *(int*)b);

}

int main()

{

   int n;

   scanf("%d",&n);

   int arr[n];

   for(int i=0;i<n;i++){

      scanf("%d",&arr[i]);

   }int min=0,max=0;

   for(int i=0;i<n;i++){

      if(arr[i]>0){
```

```
            min=arr[i];
            max=arr[i];
            break;
        }
    }
    for(int i=0;i<n;i++){
        if(arr[i]>0 && min>arr[i]){
            min=arr[i];
        }
    }
    qsort(arr,n,sizeof(int),comp);
    for(int i=0;i<n;i++){
        if(arr[i]>0 && arr[i]==min){
            min++;
        }
    }
    printf("%d",min);



    return 0;
}
```

29. Count Inversions in an Array

Description

An inversion is a pair (i,j) such that i < j and arr[i] > arr[j].

Sample Input

5

2 4 1 3 5

Sample Output

3

Explanation

Inversions are (2,1), (4,1), and (4,3).

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n,k=0;
    scanf("%d",&n);
    int arr[n];
    for(int s=0;s<n;s++){
        scanf("%d",&arr[s]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]>arr[j]){
                k++;
            }
        }
    }
    printf("%d",k);

    return 0;
}
```

30. Find Peak Element

Description

Find an element greater than its adjacent elements.

Sample Input

6

1 3 20 4 1 0

Sample Output

20

Explanation

20 is greater than both 3 and 4.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=1;i<n-1;i++){
        if(arr[i]>arr[i-1] && arr[i]>arr[i+1]){
            printf("%d",arr[i]);
            return 0;
        }
    }

    return 0;
}
```

31. Sliding Window Maximum

Description

Find the maximum element in every window of size K.

Sample Input

8

1 3 -1 -3 5 3 6 7

3

Sample Output

3 3 5 5 6 7

Explanation

Each value is the maximum of a window of size 3.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n,k;
    scanf("%d",&n);
    int arr[n];
    for(int s=0;s<n;s++){
        scanf("%d",&arr[s]);
    }
    scanf("%d",&k);
    int y=0,i;
    while(y<=n-k){
    int max=arr[y];
    for(i=y;i<y+k;i++){
        if(max<arr[i]){
            max=arr[i];
        }
```

```
    }printf("%d ",max);
    y++;

}
    return 0;
}
```

32. Maximum Circular Subarray Sum

Description

Find the maximum subarray sum considering the array as circular.

Sample Input

5

8 -8 9 -9 10

Sample Output

19

Explanation

Circular subarray [10, 8, -8, 9] gives sum 19.

33. Minimum Number of Jumps to Reach End

Description

Each element represents the maximum jump length from that position.

Sample Input

11

1 3 5 8 9 2 6 7 6 8 9

Sample Output

3

Explanation

Minimum jumps required to reach the last index are 3.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int n,k;
    scanf("%d",&n);
    int arr[n];
    for(int s=0;s<n;s++){
        scanf("%d",&arr[s]);
    }int sum=0,i=0;
    while(i<n){
        i+=arr[i];
        sum++;
    }
    printf("%d ",sum);
    return 0;
}
```

34. Find Subarray with Given Sum

Description

Find the starting and ending indices of a subarray whose sum equals K.

Sample Input

5

1 4 20 3 10

33

Sample Output

2 4

Explanation

20 + 3 + 10 = 33.

35. Chocolate Distribution Problem

Description

Distribute packets such that the difference between maximum and minimum chocolates is minimized.

Sample Input

7

7 3 2 4 9 12 56

3

Sample Output

2

Explanation

Choosing packets [2,3,4] gives difference 2.

36. Find the Median of Two Sorted Arrays

Description

Find the median of two sorted arrays.

Sample Input

3

1 3 5

3

2 4 6

Sample Output

3.5

Explanation

Merged array = [1,2,3,4,5,6].

Median = (3+4)/2 = 3.5.

37. Maximum Length Bitonic Subarray

Description

Find the longest subarray that first increases and then decreases.

Sample Input

8

12 4 78 90 45 23 10 5

Sample Output

7

Explanation

Bitonic subarray is [4,78,90,45,23,10,5].

38. Rainwater Harvesting Between Buildings

Description

Calculate trapped rainwater between buildings.

Sample Input

6

3 0 0 2 0 4

Sample Output

10

Explanation

Total trapped water is 10 units.

39. Find the K-th Largest Element

Description

Find the K-th largest element in the array.

Sample Input

6

7 10 4 3 20 15

3

Sample Output

10

Explanation

Sorted descending: 20,15,10,7,4,3.

3rd largest element is 10.

40. Maximum Rectangle Area in Histogram

Description

Given bar heights of a histogram, find the largest rectangular area.

Sample Input

7

6 2 5 4 5 1 6

Sample Output

12

Explanation

The maximum rectangle area is formed by bars with heights 4,5,5 resulting in area 12.